

Unit 1

Matrices

$$A = [a_{ij}]_{m \times n}$$

$i \rightarrow i^{\text{th}}$ row

$m \rightarrow$ number of rows

$j \rightarrow j^{\text{th}}$ Column

$n \rightarrow$ Column

Types

(i) Row matrix $\Rightarrow [a \ b \ c]$

(ii) Column matrix $\Rightarrow \begin{bmatrix} a \\ b \end{bmatrix}$

(iii) Null Matrix (zero) $\Rightarrow [0]$

(iv) Square $\Rightarrow m = n$

(v) Diagonal $\Rightarrow$ main diagonal element $\neq 0$

$$\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

(vi) Scalar matrix $\Rightarrow$ diagonal element constant

(vii) Unit matrix $\Rightarrow I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
(Identity)

(viii) Transpose $\Rightarrow m \Leftrightarrow n \Rightarrow A^T$

(ix) Symmetric $\Rightarrow A = A^T$

(x) Skew symmetric $\Rightarrow A = -A^T$

(xi) Trace of matrix $\Rightarrow$ sum of diagonal element
 $= \text{tr}(A)$

(xii) Triangular $\begin{cases} \rightarrow \text{Upper} \\ \rightarrow \text{Lower} \end{cases}$

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

$$\begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix}$$

(xiii) Equal matrix

$\Rightarrow$ order same

corresponding element are equal.

(xiv) Orthogonal $\Rightarrow AA^T = A^T A = I$

(xv) Idempotent $\Rightarrow A^2 = A$

(xvi) Involutory $\Rightarrow A^2 = I$

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- Algebra
- (i) Addition (Same order)
 - (ii) Subtraction
 - (iii) Multiplication

(a) Scalar $\times$ matrix $\Rightarrow kA$

(b) two matrix $\Rightarrow A \cdot B$ where $A = [a_{ij}]_{m \times n}$

$C = AB$ where $B = [b_{jk}]_{n \times p}$
 where $C = [c]_{m \times p}$

Determinant

order 2 $\Rightarrow \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \Rightarrow a_{11}a_{22} - a_{12}a_{21}$

order 3 $\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow a_{11}(a_{22}a_{33} - a_{23}a_{32}) - (a_{12}(a_{21}a_{33} - a_{31}a_{23}) + (a_{13}(a_{21}a_{32} - a_{31}a_{22}))$

Minors $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

for M_{11} (of a_{11}) $= \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$

$M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$

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Cofactor

$$a_{ij} \Rightarrow A_{ij} = (-1)^{i+j} M_{ij}$$

* Adjoint

$$\text{adj}(A) = [A_{ij}]^T_{n \times n}$$

here $A_{ij} \Rightarrow$ Cofactor matrix

Property $(\text{adj } A) A = A (\text{adj } A) = |A| I_n$

Singular matrix - $|A| = 0$ Non singular matrix - $|A| \neq 0$ Inverse of a Matrix

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

 $|A| \neq 0$ matrix non-singularProperty

$$(AB)^{-1} = B^{-1}A^{-1} \quad |(A^T)^{-1}| = (A^{-1})^T$$

Rank of a Matrix $\rho(A)$ non-negative integer, (0 to ∞)* at least one square sub-matrix of r order, Δ determinant $\neq 0$

Rank of Null matrix = 0

$$\text{Matrix } (m \times n) = \rho(A) \leq \min(m, n)$$

Echelon form

(i) a row which has all its elements zero below a row of non zero element

(ii) No. of zero before first non zero element in row $<$ no. of zeros in next row.

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Note $P(A)$ in echelon = No. of non zero rows

for echelon form $\Rightarrow$ by elementary transformations
make lower row = zero

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \Rightarrow$$

Normal form

$$I_r \Rightarrow \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} Z_r \\ 0 \end{bmatrix}$$

$$P(A) = r$$

Normal form

after getting echelon form make identity matrix
on constant by row & Column

Nullity of a Matrix for

$$AX = 0$$

rank nullity theorem:

here N_A = dimension of
space (Nullity)

$$\text{rank} + N_A = n \quad \text{ng. Column}$$

rank $\downarrow$ nullity

* System of linear Simultaneous eqs

$$AX = B$$

$$AX = B$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Coeffi matrix (Solution) Vector Column

$B = 0$
Homogeneous

$B \neq 0$

Inconsistent
 $P(A) \neq P(A:B)$

$D \neq 0$

Trivial

Solution

$$P(A) = P(A:B) = n$$

(Unique)

$D = 0$

Non-trivial

Solution

$$P(A) = P(A:B)$$

$< n$
(∞ soln)

$$P(A:B)$$

$\downarrow$

Rank of augmented
matrix

Solution methods

(i)

$$AX = B$$

$$\Rightarrow X = A^{-1}B$$

Invertible method

(ii)

Gauss elimination

A in echelon form by row transformation

Homogeneous system

$$AX = 0$$

 $|A| \neq 0 \Rightarrow \rho(A) = n \Rightarrow$ Trivial solution.all zero $x_1 = x_2 = x_3 = 0$ $|A| = 0 \Rightarrow \rho(A) = r < n \Rightarrow$ non trivial.at least one $x \neq 0$ Linear DependentLinear Independent

(10)

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(2)

 $\rho(A) = \text{no. of variable}$ $\rho(A) \neq \text{no. of variable}$ Eigen values $A = [a_{ij}]_{n \times n}$ Characteristic matrix of A $[A - \lambda I]$ $\lambda = [A - \lambda I] = \text{Characteristic polynomial of A}$ $|A - \lambda I| = 0 \Rightarrow \text{Characteristic equation}$ roots of equation $\Rightarrow$ eigen values (λ) • Sum of eigen value $= \text{tr}(A)$ • Product $= |A|$ • A & A^T same eigen value• Set of eigen value $=$ Spectrum of A

Eigen vector (Character vector)Put value of λ in equation

$$(A - \lambda I)x = 0$$

then find x here x is eigen vector of A Diagonalization of matrix $X_1, X_2, \dots, X_n$ Orthogonal

for

$$X_1^T X_2 = X_2^T X_1 = 0$$

$$A = P^{-1}LP$$

then B $\Rightarrow P^{-1}AP$ is diagonal matrix

diagonal elements are

eigen value of A Caley-Hamilton Theoremevery square matrix satisfy its own
characteristic equation

$$|A - \lambda I| = (-1)^n \lambda^n + c_1 \lambda^{n-1} + c_2 \lambda^{n-2} + \dots + c_n = 0$$

$$\text{Put } A = \lambda$$

$$(-1)^n A^n + c_1 A^{n-1} + c_2 A^{n-2} + \dots + c_n I = 0$$

 $\downarrow$ multiply by A^{-1} then divide by c_2